SCIENCE / BC GRADE 12

Dynamic equilibrium

Explain an equilibrium and predict how a system responds.

Chemistry 12

01 Equilibrium is dynamic

In a closed system at equilibrium, the forward and reverse reactions continue at equal rates. Observable concentrations stay constant, but the concentrations of reactants and products do not have to be equal.

02 Read an equilibrium expression

For aA + bB <=> cC + dD, Kc = [C]^c[D]^d / ([A]^a[B]^b), using equilibrium concentrations of gases or dissolved species. Pure solids and pure liquids are omitted. At a fixed temperature, Kc is constant for that reaction as written.

03 Predict a response

Le Chatelier's principle describes a shift that partially opposes a disturbance. Adding a reactant can favour its consumption. For gases, decreasing volume favours the side with fewer gas particles, if the sides differ. A catalyst speeds both directions and does not change Kc or the equilibrium composition.

Worked example

Change the concentration: $H_2(g) + I_2(g) <=> 2HI(g)$

$Kc = [HI]^2 / ([H_2][I_2])$. Suppose more H_2 is added at constant temperature and volume. The denominator initially increases, so the reaction quotient becomes smaller than Kc. The net reaction proceeds toward products until equilibrium is restored. Kc stays the same because temperature is unchanged.

Watch out: A shift toward products does not mean the forward reaction runs alone. Both directions still occur; their rates become equal again at the new equilibrium.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. At equilibrium, must reactant and product concentrations be equal?

2. Write Kc for $N_2(g) + 3H_2(g) <=> 2NH_3(g)$.

3. For this ammonia reaction, which direction is favoured by decreasing volume at constant temperature? Explain using gas coefficients.

Answer key

1. No. Forward and reverse rates are equal; concentrations are constant but need not be equal.

2. $K_c = [NH_3]^2 / ([N_2][H_2]^3)$. Each exponent comes from the balanced equation.

3. Toward products: the reactant side has 4 moles of gas per reaction as written, while the product side has 2. The shift favours fewer gas particles.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Chemistry 12

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